Sentiment Analysis-Based Study on Chinese Audiences Acceptance of "Friends"

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Github Repository

Introduction

Friends is a long-running sitcom that has attracted a large global audience, but cultural differences, such as plot developments, determine its acceptance in China.

This study uses a model to analyze online comments and explore the acceptance of Friends among Chinese audiences. By analyzing the emotions conveyed in the comments and key dialogue themes and phrases, it reveals some of the attitudes of Chinese viewers towards Friends.

System Design

The system is designed as a cloud-based data processing and analysis pipeline that integrates data storage, data processing, machine learning models, and visualization tools. The purpose of this system is to efficiently analyze user comments and generate meaningful insights through structured data processing and visualization.

The system begins with the data storage stage, where user comment data is stored in Azure Blob Storage. This cloud storage solution provides reliable and scalable storage for approximately 2,000 to 2,500 user comments collected from external sources.

Next, the data is transferred to MongoDB Atlas, which serves as the primary database system.

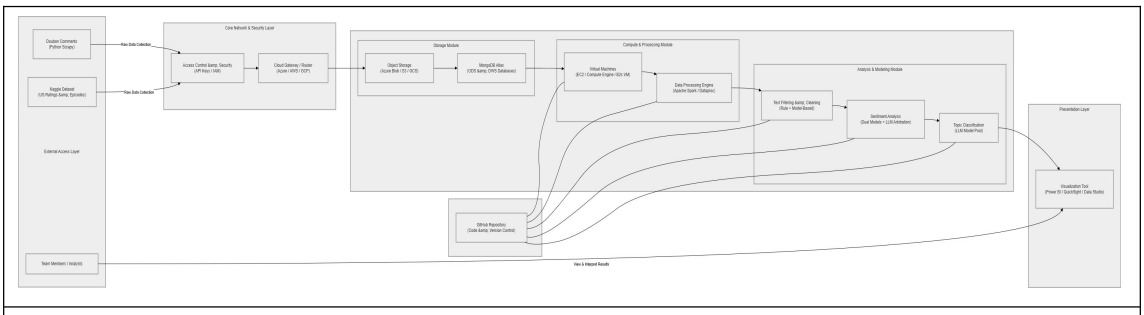
MongoDB Atlas is used to organize and manage structured datasets, allowing efficient data retrieval and processing during the analysis stage.

The data processing stage is implemented using Apache Spark, which supports large-scale data processing and improves the efficiency of data cleaning and transformation tasks. During this stage, Python is used as the primary programming language within the Google Colab cloud platform. Python scripts are applied to clean the dataset, remove duplicate records, and prepare the data for machine learning analysis.

After data processing, the system performs machine learning analysis using classification models such as Logistic Regression and Naive Bayes. These models are applied to perform sentiment analysis and categorize user comments into different emotional groups. The use of multiple models improves the reliability and accuracy of the classification results.

Finally, the processed results are exported to Microsoft Power BI, which is used as the visualization tool. Power BI generates graphical dashboards, including line charts, bar charts, pie charts, and word clouds, allowing users to easily interpret trends and patterns in user opinions.

Overall, this system design provides an efficient and scalable workflow that supports data storage, processing, analysis, and visualization in an integrated cloud-based environment.



System Architecture

Budget Plan

To evaluate the feasibility of the proposed system, three vendors were considered for development. The estimated costs included computing reources, database services, data processing, stroage, and visualization tools. All configurations are designed to support he processing of approximately 2,000-2,500 user comments.

Vendor A – Microsoft Azure

Microsoft Azure is selected as the primary cloud platform due to its strong integration with data storage and visualization tools.

Component	Service	Monthly Cost(HKD)
Virtual Machine	Azure B2s VM(2vCPU,4GB RAM)	800
Database	MongoDB Atlas	600
Data Processing	Azure Spark	800
Storage	Azure Blob Storage	200
Visualization	Power BI	100

Total Monthly Cost:HKD2,700

Total Yearly Cost:HKD32,400

The total cost is well below the HKD 200,000 budget limit and is sufficient to process approximately 2,285 user comments.

Vendor B – Amazon AWS

Component	Service	Monthly Cost(HKD)
Virtual Machine	Amazon EC2 t3.medium	900
Database	AWS	600
Data Processing	Spark	900
Storage	Amazon S3	250
Visualization	AWS QucikSight	150

Total Monthly Cost:HKD2,800

Total Yearly Cost:HKD33,600

The AWS-based system provides similar computing capability and supports large-scale data processing using Spark. The total yearly cost remains significantly below the HKD 200,000 budget limit.

Vendor C – Google Cloud

Component	Service	Monthly Cost(HKD)
Virtual Machine	Google Compute Engine	850
Database	MongoDB Atlas	600
Data Processing	Google Dataproc	850
Storage	Google Cloud Storage	200
Visualization	Google Data Studio	120

Total Monthly Cost:HKD2,620

Total Yearly Cost:HKD31,440

The GCP-based configuration offers a cost-effective solution with strong integration for cloud-based data processing and visualization tasks.

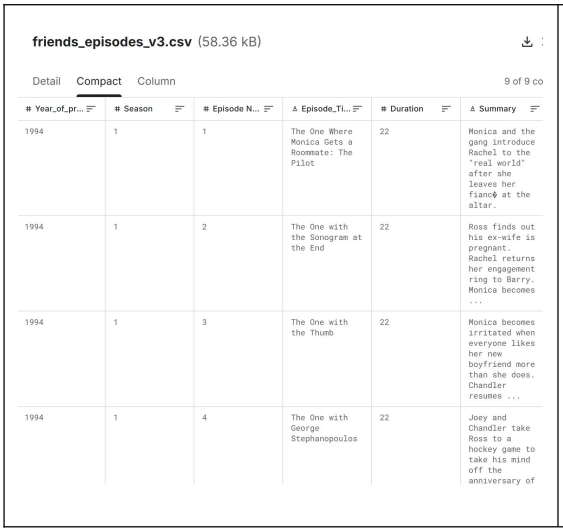
Method

We haved devided the whole into 4 steps, which are Data collect, Data Extract, Make Model, and Data Virtualization.

1. Data Collect (ODS->DWS)

1.1. Data Source

We use douban platform as our data source. Douban as most popular and television review community in China. We through this platform to scrapy chinese users short comments and rating. And then, in order to compare with the rating between China and America, we get some csv files from Kaggle.

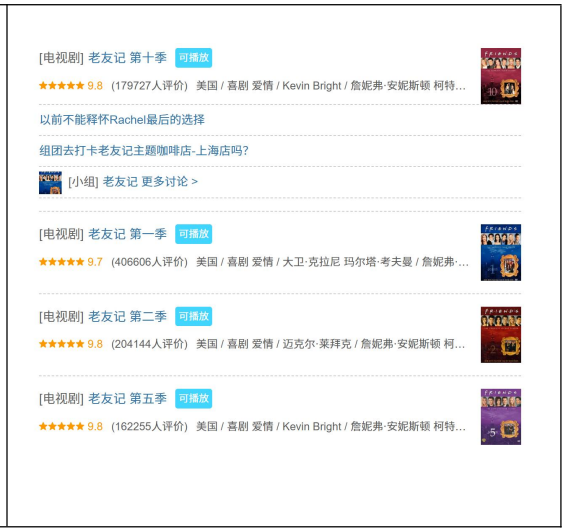


friends_episodes_v3.csv (58.36 kB)

Detail Compact Column 9 of 9 co

# Year_of_pr...	# Season	# Episode N...	A Episode_Ti...	# Duration	A Summary
1994	1	1	The One Where Monica Gets a Roommate: The Pilot	22	Monica and the gang introduce Rachel to the "real world" after she leaves her fianc◆ at the altar.
1994	1	2	The One with the Sonogram at the End	22	Ross finds out his ex-wife is pregnant. Rachel returns her engagement ring to Barry. Monica becomes ...
1994	1	3	The One with the Thumb	22	Monica becomes irritated when everyone likes her new boyfriend more than she does. Chandler resumes ...
1994	1	4	The One with George Stephanopoulos	22	Joey and Chandler take Ross to a hockey game to take his mind off the anniversary of

Kaggle data file



[电视剧] 老友记 第十季 可播放

9.8 (179727人评价) 美国 / 喜剧 爱情 / Kevin Bright / 詹妮弗·安妮斯顿 柯特...

以前不能释怀Rachel最后的选择

组团去打卡老友记主题咖啡店-上海店吗?

[小组] 老友记 更多讨论 >

[电视剧] 老友记 第一季 可播放

9.7 (406606人评价) 美国 / 喜剧 爱情 / 大卫·克拉尼 玛尔塔·考夫曼 / 詹妮弗·...

[电视剧] 老友记 第二季 可播放

9.8 (204144人评价) 美国 / 喜剧 爱情 / 迈克尔·莱拜克 / 詹妮弗·安妮斯顿 柯...

[电视剧] 老友记 第五季 可播放

9.8 (162255人评价) 美国 / 喜剧 爱情 / Kevin Bright / 詹妮弗·安妮斯顿 柯特...

Douban platform

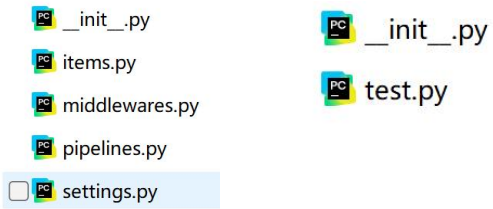
1.2. Acquisition Technology

We use Python scrapy to collect data from Douban platform. We used Python Scrapy as our crawling framework. The basic process of Python Scrapy have 4 steps.

Operation code	Examples of pictures

scrapy startproject+"the name of scrapy file"	`C:\Users\任钱\Desktop>scrapy startproject douban`
	`New Scrapy project 'douban', using template directory`
	`ckages\scrapy\templates\project', created in:`
	`C:\Users\任钱\Desktop\douban`
	
	`You can start your first spider with:`
	`cd douban`
	`scrapy genspider example example.com`

scrapy genspider+"scrapy name"+"crawler website"	`C:\Users\任钱\Desktop\douban>scrapy genspider test www.douban.com`
	`Created spider 'test' using template 'basic' in module:`
	`douban.spiders.test`

Python file edit	
scrapy crawl+"scrapy name"	`C:\Users\任钱\Desktop\douban>scrapy crawl test -o data.csv`

There also have more some detail about Python Scrapy.

Regarding 'Python file edit', the different Python file have different function. There we only talk some

basic files.

1) item.py: This file is used to set data schema. It defined data schema and data type so that provide uniform structural standard for data collect, data cleaning and storage.

2) settings.py: This file is used to set project configuration.

3) test.py: This is core crawler implementation file. You need to edit code to achieve crawl data from website to the specified file.

And then talk about scrapy output command.

scrapy crawl+"scrapy name": Look at picture, the '-o data.csv' represent that file name and the type of file. If you want to output json file, you can 'scrapy crawl test -o filename.json'.

1.3.Data Schema & Storage

1) Data Schema

About the data of douban platform, we design 2 tables, a 'ods_douban_comment table' and 'ods_douban_season_rating'. So they have different data schema.

Table 1: ods-douban-comment Fields: id season (int) comment (String) name (String)	Table 2: ods _ douban _ season _ rating Fields: season (string) rating (double) total _ people (int)

ods_douban _comment table ods_douban_season_rating table

Regarding the data that come from Kaggle, since they are already well organized and public accessible, so we preserved the orginal data structure and import data into a table that called 'ods_friends_episodes_v3'.

Table3: ods _ friends _ episodes _ v3
Fields: Year-of-prod (int)
Season (int)
Episode Number (int)
Episode _ Title (String)
Summary (String)
Director (String)
Rating (double)
Votes (int)

ods_friends_episodes_v3 table

2) Data Storage

We stored the collected data in the MongoDB-based ODS database. The full name of ODS is

'Operational Data Store'. This layer is used for temporary storage layer and construct a connection between raw data sources and data warehouse.

1.4.Data Size

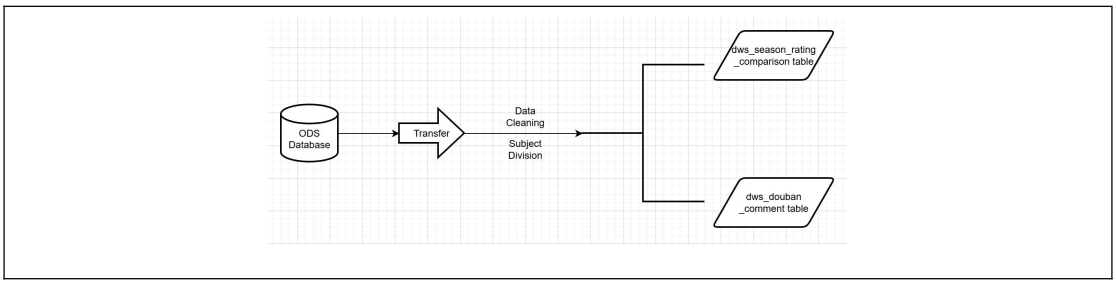
We use Python Scrapy to scrapy 4000 short comments from douban ,then output as CSV file. And we get 10 rating data of chinese audiences. Also we get 10 rating data of American audiences.

2. Data Extract

2.1. Process

Focus on Data Extract, we followed the path that from ODS to ETL process, and eventually to the DWS layer.

The full name of ETL is 'Extract, Transform, Laod'. The process that combines data from mutiple data sources into a single. In our project, we extracted some important data and loaded into MongoDB-based DWS database. We built this database through topic aggregation and data cleaning based on the ODS layer data.



The workflow of Data Extract

2.2. Detail

1) Data cleaning

In this project, during the ETL process from the ODS to the DWS layer, we first perform data deduplication, followed by sorting the records in ascending order based on the TV series season.

2) Subject divison

Based on the field topic of ODS layer, we divide and summarize the data into two core topic tables: rating comparison table and comment table. The first table aggregates rating indicators of each season to quantify audience feedback, and the second table aggregates audience comments of each season. After data cleaning and unifying the indicator caliber, we finally form a standardized and efficient indicator query service in DWS layer, providing solid data support for modeling and analyzing the acceptance of friends in China in ads layer.

3) Some code

In order to improve the code readability and data search effeciency, we use some sql statement.

```
#生成 dws_season_rating_comparison表
df_dws_rating=duckdb.query("""
  SELECT d.season,
      d.rating AS df_douban_rating,
      i.imdb_rating,
      ROUND((d.rating-i.imdb_rating),1) AS rating_difference
  FROM df_ods_douban_rating d
  LEFT JOIN (SELECT ROUND(AVG(Stars),1) AS imdb_rating,
             Season
         FROM df_ods_imdb_rating
         GROUP BY Season) i
  ON d.season=i.Season
  ORDER BY d.season
""").df()
```

```
#生成 dws_douban_comment
df_dws_comment=duckdb.query("""
  SELECT id AS user_id,
      season,
      "name" AS user_name,
      comment
  FROM df_ods_comment
  ORDER BY season
""").df()
```

2.3. Subject Table Construction

We constructed 2 subject table: a 'dws_season_rating_comparison' table and a 'dws_ douban_comment' table. They have different function.

1) dws_season_rating_comparison: Aggregates rating indicators of each season to quantify audience feedback.

2) dws_ douban_comment: Aggregates audience comments of each season.

Table 1. dws_season_rating_comparison Fields: season (int) douban-rating (double) imdb-rating (double) rating-difference (double)	Table 2: dws_douban_comment Fields: user_id name (String) comment (String) season (int)

dws_season_rating_comparison table　　dws_douban_comment table

3. Make Model

Make Model in this project, is the core analysis model. And it is called ADS. ADS is the Application Data Store, which enable data to quickly serve specific application scenarios in a business-friendly form. In this project, its purpose is use to

It has 7 steps to make sure that the source data is clean, and put the raw comment to the accurate, structured, standardized results:

DWS Comment Raw Data
STEP1: Rule-Based Filtering
STEP2: Model-Based Filtering
STEP3: Text Processing
STEP4: Sentiment Analysis
STEP5: Topic Classification
STEP6: Merge Sentiment And Topic Results
STEP7: Output Results

*I-M-3 1　Resume Model Overall Seven-Step Flowchart

3.1 Rule-Based Filtering

Before the sentiment analysis, to improve the data quality, we filtering the raw douban data with rule. This step include delete the null comment, standardize text formatting, remove meaningless comments

that only contain symbols or numbers, and delete duplicate comments.

After all these steps, the data become more clean, and suitable for analysis. It will output a new table called dws_comment_rule_filtered, the raw comments number is 4235, and output comments becomes 3781.

3.2 Model-Based Filtering

After the rules filter, we use the model to recognize the garbage comment, to reduce influence of spam, attack and meaningless content. Specifically, we combine the output of multi-text classify model, and get the scores by calculated using a weighted average to improve overall stability.

The Score formula of the single model is :

$$\text{Score}_n = \begin{cases} CE, & \text{if Tag is Regular} \\ 1 - CE, & \text{otherwise} \end{cases}$$

*CE: Coefficient of Normal tag

*I-M-3-2 2 Score formula of the model

The final score formula is:

$$\text{Score}_{\text{final}} = \text{Score}_{\text{sec}} \times 0.2 + \text{Score}_{\text{tir}} \times 0.35 + \text{Score}_{\text{fou}} \times 0.45$$

*I-M-3-2 3 Final formula of comment

Only if the final score bigger or equal than 0.6 can get into the next step.

After the model was cleaned up, the number of comments changed from 3781 to 2642 after the rule was cleaned up.

3.3 Text Pre-Processing

In this step includes segmenting the text into sentences, words, removing stop words, and deleting sentences that are too short or lack sufficient information. Through handled by this step, the text of comment are more standardized, and more suitable for the keyword extraction.

3.4 Sentiment Analysis And Keyword Extraction

This is the core step of ADS, through which we obtain sentiment and keywords. In this step, we combine multiple sentiment analysis models to judge comments and introduce an LLM Model to handle conflicting results, thereby improving accuracy and stability.

3.4.1 Sentiment Analysis

In sentiments analysis, we use two local sentiment analysis models to jointly judge comments.

The first model will output a 5-star scale, we map these values to a unified sentiment score for subsequent analysis. The score mapping method is as follows:

$$P(\text{sentiment}|\text{star}=s)=\begin{cases}\mathbf{N}=1,\mathbf{Ne}=0,\mathbf{P}=0 & s=1\\ \mathbf{N}=0.7,\mathbf{Ne}=0.3,\mathbf{P}=0 & s=2\\ \mathbf{N}=0,\mathbf{Ne}=1,\mathbf{P}=0 & s=3\\ \mathbf{N}=0,\mathbf{Ne}=0.3,\mathbf{P}=0.7 & s=4\\ \mathbf{N}=0,\mathbf{Ne}=0,\mathbf{P}=1 & s=5\end{cases}$$

*I-M-3-4-1 2 5 Star model score mapping

The second model will output the positive, negative, neutral probability of each comments. Cause this model has limited ability on long sentience, we need to use the step 3.3 (Text Pre-processing) result as input. It will compute the sentience sentiment, then the entire comment is weighted and summarized based on sentence length. The output sentience probability like this:

$$\mathbf{p}^{(i)}=\begin{bmatrix}p_{\mathbf{N}}^{(i)}\\ p_{\mathbf{Ne}}^{(i)}\\ p_{\mathbf{P}}^{(i)}\end{bmatrix},\quad \text{s.t.}\quad p_{\mathbf{N}}^{(i)}+p_{\mathbf{Ne}}^{(i)}+p_{\mathbf{P}}^{(i)}=1$$

*I-M-3-4-1 3 3 Classification model Single Sentences Output

After getting the whole sentience of the comments, we further summarized the sentiment index of the entire comment. The calculation method is as follows:

$$\mathbf{P}_{\text{tri}}=\mathbf{W}^{\top}\mathbf{P}_{\text{sent}}=[w_1\quad w_2\quad\cdots\quad w_n]\begin{bmatrix}p_{\mathbf{N}}^{(1)} & p_{\mathbf{Ne}}^{(1)} & p_{\mathbf{P}}^{(1)}\\ p_{\mathbf{N}}^{(2)} & p_{\mathbf{Ne}}^{(2)} & p_{\mathbf{P}}^{(2)}\\ \vdots & \vdots & \vdots\\ p_{\mathbf{N}}^{(n)} & p_{\mathbf{Ne}}^{(n)} & p_{\mathbf{P}}^{(n)}\end{bmatrix}\qquad w_i=\frac{l_i}{L}=\frac{l_i}{\sum_{j=1}^{n}l_j}$$

*Set comment number is n, the length of sentence n_i is l_i, the comment whole length is $L=\sum_{\mathrm{j}=1}^{\mathrm{i}}l_i$

*I-M-3-4-1 4 Comment coefficient output by 3 classification model output

Then, we will also incorporated domain-specific vocabulary to appropriately refine the model results.

This will reduce the number of cases where some obvious sentiment words are not fully recognized by the model. After finish the correction, we will get into the final decision making according to the model result. if the coefficient of two model both too low, we judge this comment as neutral, if the two results clearly in conflict, we will introduced a LLM model to arbitration. In other cases, we obtain the result by through normal weighting. The merging formula is as follows:

$$P(\text{Negative}) = 0.6 \times P_A(\text{Negative}) + 0.4 \times P_B(\text{Negative})$$

*I-M-3-4-1 5 Weighting formula for combining the results of two sentiment models

To improve the reasonableness of the results, we added a direction protection mechanism to avoid fusion results caused by inconsistencies in judgments between patterns. Finally, each comment will be divided into three categories: positive, negative, and neutral. Therefore, even negative results will be sent to LLM for further verification, thereby further reducing the risk of misjudgment.

3.4.2 Keyword Extraction

In the topic extraction, we used a layered extraction method to improve the stability and effectiveness. First, we use keyword extraction model to generate candidate keywords or phrases from the original comments, the most representative results are selected as initial candidates. If the model fails to extract valid keywords, it reverts to the original comment text and continues subsequent processing.

Then rule-based cleaning is used to retain terms of moderate length, without spaces, and not consisting solely of symbols or numbers, in order to reduce noise interference and obtain a cleaner set of candidate words.

After handle all of these steps, if it still exists available keywords, further extraction and filtering were conducted to select keywords that better summarize the core meaning of the comments. Through these way, we can extract the more stable and more representative keywords.

3.5 Topic Classification

In this stage, we use LLM Models to classify the topic. Considering that the limit of stability and availability on single Model, we build a models pool which contain multiple Model to classify the topic,

improve the efficiency.

Before the formal use, we will test all Models in the pool, just reserve the model with high stability. The comments will assigned to different models for classification, if a model fails, it will switch to another model. If all models fail, this comment will be classify as "others" categories.

During the classification, Models will determine the topic and return the tags and reasons. After all these steps, every comment could get the topic categories.

3.6 Merge Sentiment And Topic Results

In this stage, we will further integrate the two sets of results into a comprehensive analysis table, thereby achieving a joint presentation of comment sentiment and comment topic.

This step uses the sentiment analysis results table and the topic classification results table as input. To ensure the quality of the merging, the system will first reduplicate the topic classification results according to the season and content of the comment, retaining the last result in the duplicate records. Subsequently, the sentiment analysis results, as the main body, are merged with the topic classification results using a left join, so that each comment retains its original sentiment analysis information while supplementing it with the corresponding topic category.

3.7 Output Results

After finish the main analysis pipeline, the system will continue output the two application results, which are the comments table and seasons table. Comments table is used to keep the core analysis information, and the Season table is used to summarize the information of different season.

3.7.1 Comment-Level Results Table

The comment result table is used to show the core analysis conclusion. This table keep the key fields, which are {season, comment, keyword, sentiments tags, topic class}, so it can reflect about the comments, what it says, and what is its sentiments.

3.7.2 Quarterly Summary Table

We further constructed a quarterly summary results table to reflect audience feedback across different quarters from an overall perspective. This table statistically analyzes the number of positive, negative, and neutral reviews quarterly, and incorporates external rating data, including Douban ratings, IMDb ratings, and the difference between the two. By combining platform ratings with the sentiment distribution of reviews, this results table provides a more comprehensive reflection of the overall performance of works across different quarters.

4. Visualization

After completing the sentiment analysis and topic classification process, the results were presented through data visualization to provide a more intuitive understanding of audience perspectives and discussion trends. Based on the processed dataset, this project utilized Power BI to generate visualizations. This stage primarily focused on rating comparisons, topic distribution, sentiment distribution, and keyword frequency analysis, allowing users to efficiently interpret the analytical results.

The first category of visualization consists of a line chart, which compares ratings from different platforms. The results show that ratings on Douban are generally higher than those on IMDb. This comparison helps illustrate differences in audience perception between Chinese and international users, indicating that Chinese audiences tend to give slightly higher ratings to the TV series *Friends*.

The second category of visualizations consists of bar charts, which illustrate the distribution of various discussion topics extracted from user comments. The results show that plot-related topics account for the largest number of comments, followed by nostalgia-related content, and then actor-related discussions. This indicates that audiences are primarily concerned with plot development, while emotional resonance and actor performance also play significant roles in audience engagement.

The third category of visualizations consists of pie charts, which present the distribution of sentiment categories across all comments. The charts show that the majority of comments express positive sentiment, while a smaller portion are neutral, and only a very small proportion are negative. This distribution reflects an overall positive attitude among audiences toward the series.

The final category of visualizations consists of a word cloud, which displays high-frequency keywords

extracted from audience comments. Frequently appearing words such as “like,” “friends,” and “touching” highlight the emotional and interpersonal dimensions valued by viewers. These keywords suggest that themes such as friendship, emotional connection, and nostalgic memories have a strong influence on audience perceptions of the series.

Overall, the visualizations presented above provide clear and intuitive support for the analytical findings of this project. By transforming the processed textual data into graphical representations, the visualization stage enhances the readability of the results and effectively demonstrates that Friends continues to maintain a relatively high level of acceptance among Chinese audiences.

Result

In conclusion, through these visualizations, we can confidently conclude that Friends enjoys a high acceptance rate in China, characterized by high ratings, positive sentiment. According to graphs, we can find that words like' Classic, 'Like', and some character names appear largest. That also means Chinese viewers connect deeply with the characters and view the show as a classic.

Group Contribution

Hu Jihan : Make model for text analysis

Chen Jiahao: Data Collect and Data Extraction

Tang Chao: Data Visualization and System

GitHub Repository

Project GitHub Repository [*https://github.com/mgHurryo/STAT_2630SEF_Group_Project*]

References

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4. GitHub, for open-source NLP libraries including jieba, KeyBERT, and SnowNLP; and stopwords-zh liabrary.
5. DeepSeek API, for LLM-based arbitration in sentiment analysis.
6. AliYun API, for LLM-based topic classification and model pool support.
7. Moonshot API, for supplementary LLM-based topic classification.